

## 4.10 Castle Forbes Bay

### Catchment description

Castle Forbes Bay Rivulet (catchment area = 7 km<sup>2</sup>) runs off the foothills in the western catchment, travelling only 11 km to the Huon Estuary. Much of the upper catchment is forested, with land cleared in the middle and lower sections for grazing, plantation forestry, orchards and rural residences. During 1997-98, Castle Forbes Bay Landcare removed most of the streamside crack willows and replanted the zone with native species.

### Monitoring program

Water temperature, turbidity, conductivity, pH, faecal bacteria and ortho-phosphate were measured on between two and thirty-one occasions at three sites along the catchment (Table 4.10.A).

**Table 4.10.A** Castle Forbes Bay Rivulet monitoring sites.

| Sites  | Site location    | Tested by                    | Times/year | Start | Finish  |
|--------|------------------|------------------------------|------------|-------|---------|
| CFB010 | Top of CFB Rd    | Castle Forbes Bay Waterwatch | 12         | 1997  | Current |
| CFB020 | End of sealed rd | Castle Forbes Bay Waterwatch | 12         | 1997  | Current |
| CFB030 | Below weir       | Castle Forbes Bay Waterwatch | 12         | 1997  | Current |

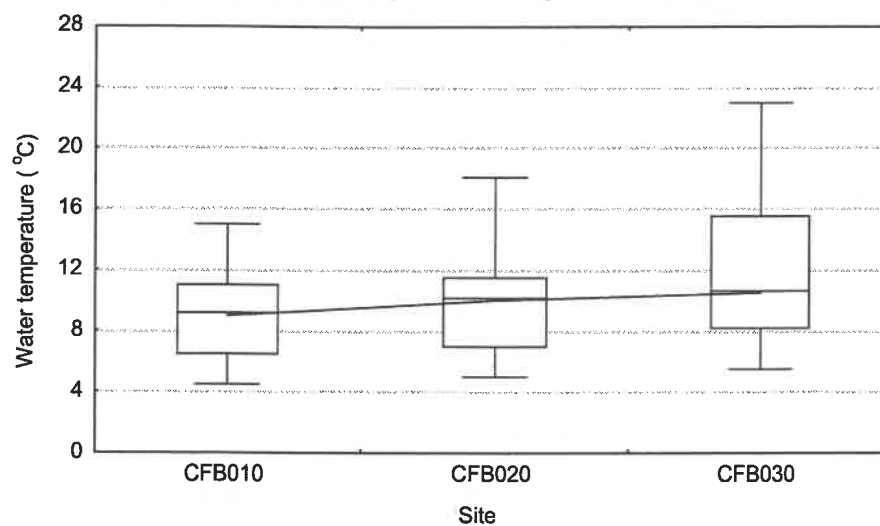
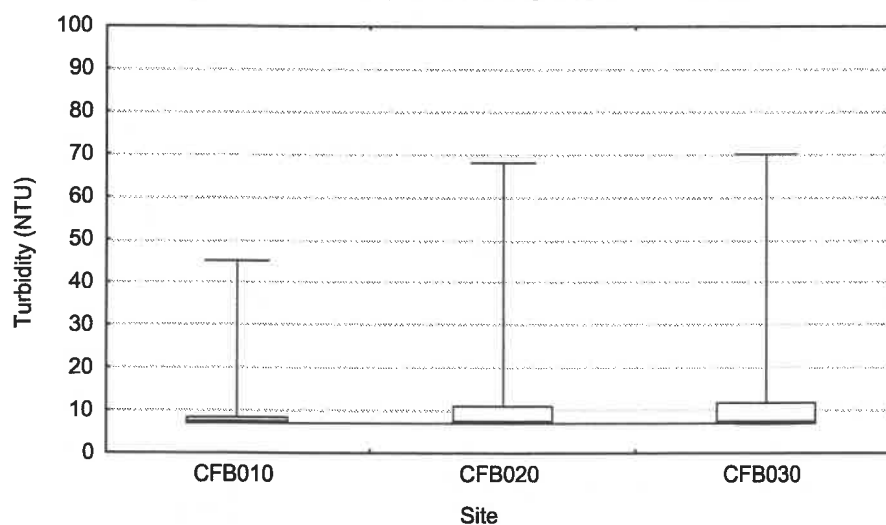
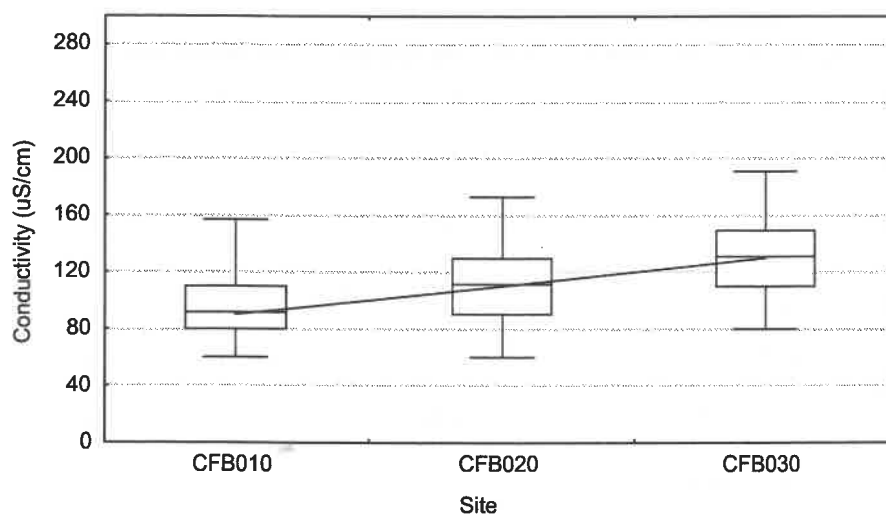
### Results and Discussion

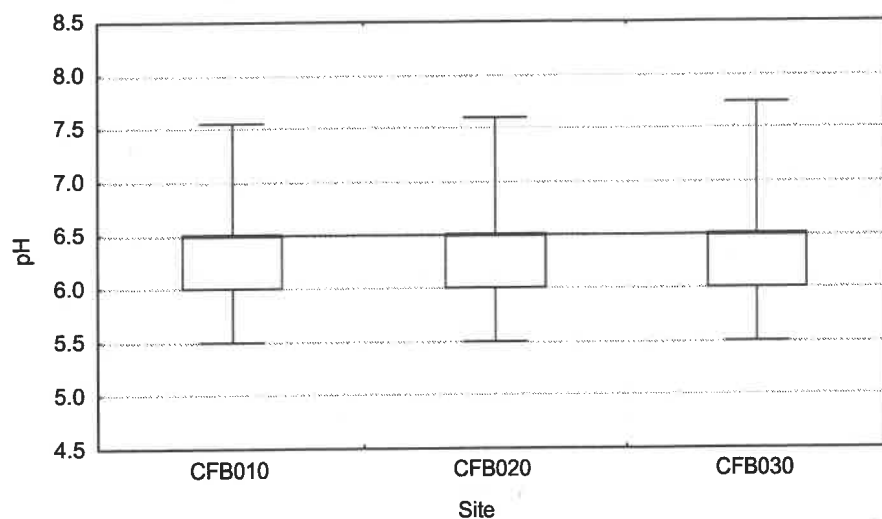
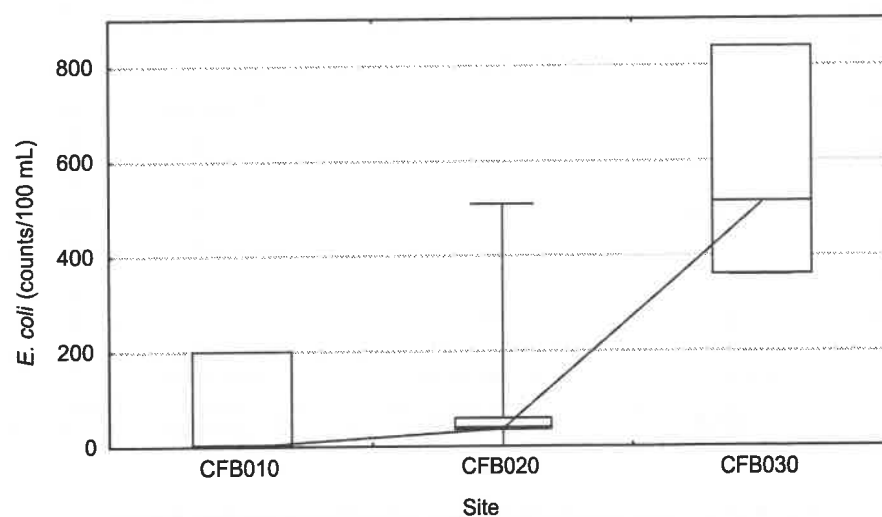
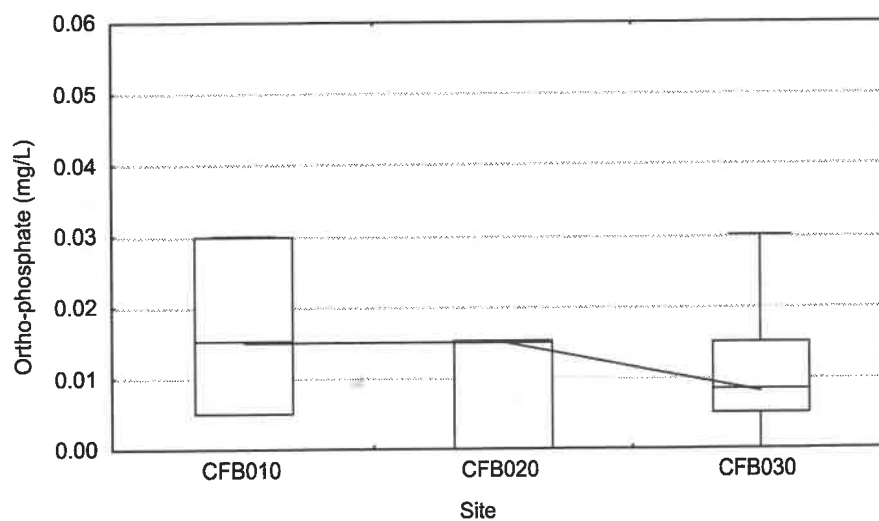
Median water temperatures at the three sites were between 9 and 11 °C, despite differences in the amount of riparian vegetation directly upstream (Fig. 4.10.A). There was a wide seasonal variation of between 5 and 23 °C, which is likely to be in response to air temperatures since the rivulet has a small volume.

Median turbidity levels in Castle Forbes Bay Rivulet were very low, at less than 7 NTU, although very elevated peaks (above 45 NTU) occurred during and up to 36 hours after heavy rain (Fig. 4.10.B). Conductivity levels were also very low and the median level increased only very slightly down the catchment (Fig. 4.10.C). Median pH values were slightly acidic across the catchment, which is normal (Fig. 4.10.D).

Median *E. coli* counts were low at the top and middle sites (below 100 counts/100 mL) compared to the bottom site (675 counts/100 mL) (Fig. 4.10.E). However, middle and bottom sites recorded significantly higher peaks during some summer months, indicating a health risk for primary contact activities during this period. Sources of the faecal bacteria were likely to include ineffective septic systems and uncontrolled stock access to the rivulet.

Low phosphate levels of between 0.015 and 0.03 mg/L were recorded at the three sites (Fig. 4.10.F). The higher levels may well be natural peaks due to the breakdown of organic material such as willow leaves and algae.

**Fig. 4.10.A Water temperature along Castle Forbes Bay Rivulet.****Fig. 4.10.B Turbidity levels along Castle Forbes Bay Rivulet.****Fig. 4.10.C Conductivity levels along Castle Forbes Bay Rivulet.**

**Fig. 4.10.D pH levels along Castle Forbes Bay Rivulet.****Fig. 4.10.E *E. coli* levels along Castle Forbes Bay Rivulet.****Fig. 4.10.F Ortho-phosphate levels along Castle Forbes Bay Rivulet.**

### Recommendations

The water quality information suggests that within the catchment there needs to be better management of septic systems and domestic animals near waterways. Other possible land use problems include vegetation management and other forms of soil disturbance (Table 4.10.B).

The current revegetation of riparian zones along Castle Forbes Bay Rivulet should be continued and accelerated in some areas. Before the removal of any of the remaining willows, the issues of shading, erosion and stock access should be addressed.

The monitoring program undertaken by Castle Forbes Bay Landcare should continue, with more regular *E. coli* and ortho-phosphate testing. Castle Forbes Bay Landcare should continue to receive the support of a Waterwatch network.

**Table 4.10.B** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall |
|------------|-----------|--------------|-----------|-----------------|----------------|---------|
| Rating     | 4         | 3            | 4         | 3               | 2              | 16      |
| Assessment | Excellent | Good         | Excellent | Good            | Fair           | Good    |

## 4.11 Dover Rivulet

### Catchment description

Dover Rivulet is a small creek running over 8 km predominantly through low scale agricultural activities including orchards and livestock grazing. There are many stands of crack willow along the creek and few intact natural riparian zones. The lower section of Dover Rivulet flows through the Dover town area. Dover Rivulet forms a small lagoon at its mouth, which is periodically flushed by high tides in Port Esperance.

### Monitoring program

Water temperature, turbidity, conductivity and pH were sampled on between two and four occasions at five sites along the catchment (Table 4.11.A). Ortho-phosphate and *E. coli* levels were measured on two occasions at the top site.

**Table 4.11.A** Dover Rivulet monitoring sites.

| Sites  | Site location      | Tested by           | Times/year | Start | Finish  |
|--------|--------------------|---------------------|------------|-------|---------|
| DOV070 | Huon Highway       | Bobbi (1998)        | 2          | 1997  | 1997    |
|        |                    | Dover District High | 1          | 1998  | 1998    |
|        |                    | Healthy Rivers      | Irregular  | 1999  | Current |
| DOV090 | Reeves Lane        | Dover District High | 1          | 1998  | 1998    |
|        |                    | Healthy Rivers      | Irregular  | 1999  | Current |
| DOV110 | Southgate Market   | Dover District High | 1          | 1998  | 1998    |
|        |                    | Healthy Rivers      | Irregular  | 1999  | Current |
| DOV120 | Chapman Ave        | Dover District High | 1          | 1998  | 1998    |
|        |                    | Healthy Rivers      | Irregular  | 1999  | Current |
| DOV130 | Esperance Coast Rd | Dover District High | 1          | 1998  | 1998    |
|        |                    | Healthy Rivers      | Irregular  | 1999  | Current |

### Results and Discussion

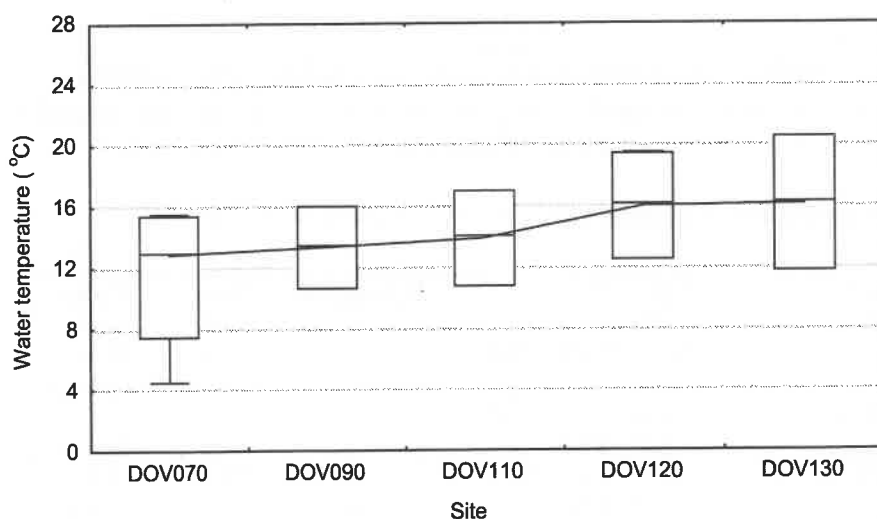
Dover Rivulet has moderately high water temperatures, with the median values at the five sites varying between 13 and 16 °C (Fig. 4.11.A). The higher water temperature is most likely to be due to the lack of significant riparian vegetation above most of the monitoring sites. However, it should be noted that little monitoring occurred during winter months.

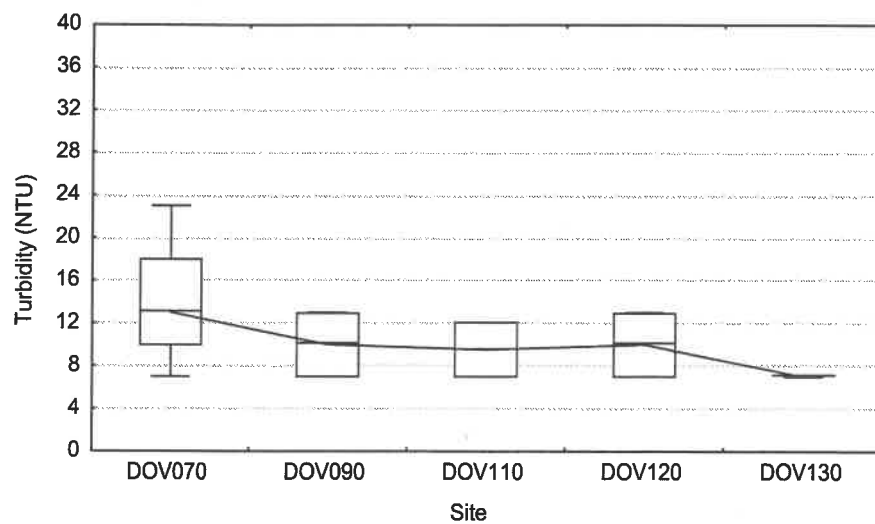
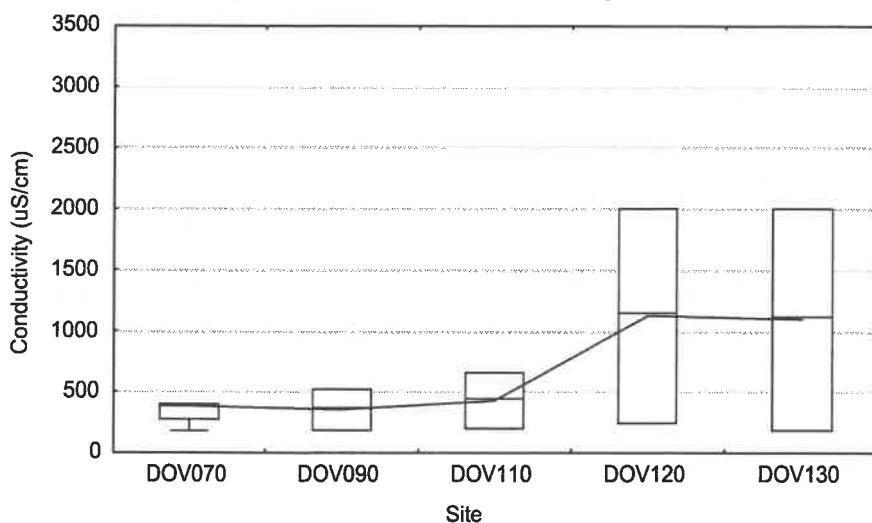
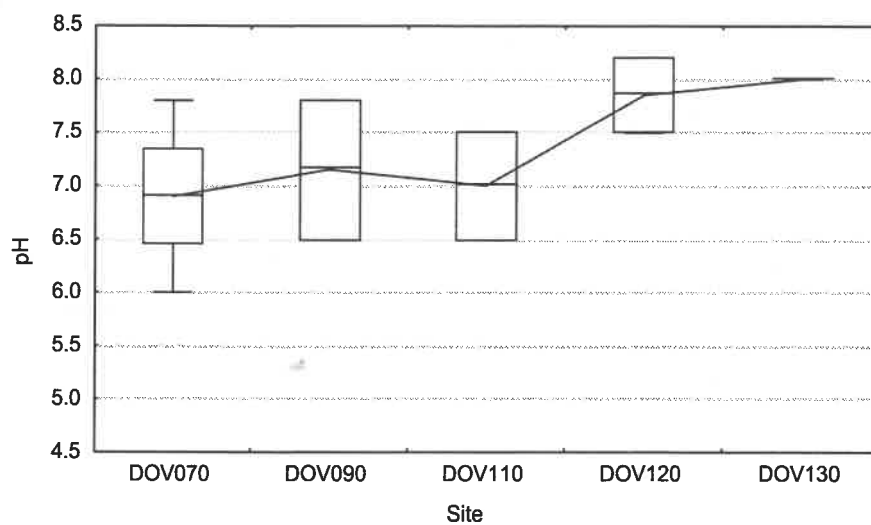
The median turbidity level decreased down the catchment, from 13 NTU at the top site to less than 7 NTU at the bottom site (Fig. 4.11.B). This trend indicates significant soil and riverbank erosion in the upper sites, perhaps due in part to unrestricted cattle access to waterways and the creek being forced to erode new pathways due to the extensive willow stands blocking flow through the natural creek bed. The lower turbidity levels in the bottom of the catchment indicated that there was less sediment input in this part of the catchment and the slower flow rate allowed the sediment to settle out before the creek reached the Huon Estuary.

Conductivity levels in Dover Rivulet are moderately high, with median levels at all five sites close to 400  $\mu\text{S}/\text{cm}$  (Fig. 4.11.C). The two bottom sites, with median levels close to 1000  $\mu\text{S}/\text{cm}$ , were within the tidal limit. The influence of the salt wedge was also reflected in the pH levels, with neutral pH levels in the middle of the catchment and more alkaline pH levels at the two lower sites (Fig. 4.11.D).

Two ortho-phosphate and *E. coli* snapshot tests were conducted at DOV070 in 1997, with slightly elevated levels of 0.006 and 0.008 mg/L and 94 and 270 counts/100 mL respectively recorded (Bobbi, 1998).

Fig. 4.11.A Water temperature along Dover Rivulet.



**Fig. 4.11.B Turbidity levels along Dover Rivulet.****Fig. 4.11.C Conductivity levels along Dover Rivulet.****Fig. 4.11.D pH levels along Dover Rivulet.**

### Recommendations

More water quality data and a comprehensive catchment survey are required to form a complete picture about the health of the catchment. The water quality information suggests within the catchment there needs better management of septic systems and domestic animals near waterways and vegetation, and other forms of soil disturbance (Table 4.11.B). Rehabilitation of riparian zones is required along much of the creek to improve shading.

Crack willow trees should be removed where possible to reduce stream width, stream erosion and allow better stock management. Before any willow removal, the issues of shading, erosion and stock access need to be addressed.

A regular water quality-monitoring program should be implemented at Dover Rivulet to monitor all parameters, particularly faecal bacteria and turbidity levels.

**Table 4.11.B** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall |
|------------|-----------|--------------|-----------|-----------------|----------------|---------|
| Rating     | 3         | 2            | 4         | 3               | 2              | 14      |
| Assessment | Good      | Fair         | Excellent | Good            | Fair           | Good    |

## 4.12 Esperance River

### Catchment description

The Esperance River (catchment area = 175 km<sup>2</sup>) originates in the Hartz Mountains National Park. Much of the upper and middle catchment is located in State Forest, including reserved land, *Eucalyptus* plantations and timber harvesting areas. Esperance River follows down through the old townships of Raminea and Strathblane before entering Port Esperance. Dover's water supply is provided by the Esperance River.

### Monitoring program

Water temperature, turbidity, conductivity and pH were tested on 90 occasions between 1997 and 2000 at two sites (Table 4.12.A).

**Table 4.12.A** Esperance River monitoring sites.

| Sites  | Site location                                         | Tested by         | Times/year | Start | Finish  |
|--------|-------------------------------------------------------|-------------------|------------|-------|---------|
| ESP001 | Esperance River Rd<br>(above Forestry Tasmania)       | Forestry Tasmania | 26         | 1997  | Current |
| ESP002 | Above Dover water intake<br>(below Forestry Tasmania) | Forestry Tasmania | 26         | 1997  | Current |

### Results and Discussion

Meyer and Lynch (2000) results showed a flow of excellent quality water down through the catchment. While forestry activities do have the potential to effect water quality, the volume in Arve River may dilute any effects from increases in soil erosion, removal of vegetation and use of agricultural chemicals.

### Recommendations

The water quality information suggests that the catchment is not significantly effected by the present land uses and the current land and water management practices currently in place should be continued (Table 4.12.B). The current monitoring program should be continued. Agricultural chemical monitoring could be included if warranted.

**Table 4.12.B** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall   |
|------------|-----------|--------------|-----------|-----------------|----------------|-----------|
| Rating     | 4         | 3            | 4         | Not monitored   | Not monitored  | 18        |
| Assessment | Excellent | Good         | Excellent |                 |                | Excellent |

## 4.13 Fluertys Creek

### Catchment description

Fluertys Creek (catchment area = 11 km<sup>2</sup>) runs off the foothills in the western section of the catchment, travelling only 4 km to the Huon Estuary. Much of the catchment is forested, with rural residences and a variety of low scale agriculture activities lower in the catchment.

### Monitoring program

Water temperature, turbidity, conductivity, dissolved oxygen, ortho-phosphate and pH were tested on two occasions at one site (Table 4.23.A).

**Table 4.23.A** Fluertys Rivulet monitoring sites.

| Sites  | Site location | Tested by    | Times/year | Start | Finish |
|--------|---------------|--------------|------------|-------|--------|
| FLU100 | Huon Highway  | Bobbi (1998) | 2          | 1997  | 1997   |

### Results and Discussion

The limited testing indicated that the water quality was good, with low turbidity and nutrient levels, slightly elevated conductivity and faecal bacteria levels and high levels of dissolved oxygen (Bobbi, 1998).

### Recommendations

Significantly more water quality data is required to form a complete picture about the health of the catchment. The water quality information suggests within the catchment there needs better management of septic systems and domestic animals near waterways and vegetation, particularly on hills and near waterways, and other forms of soil disturbance (Table 4.23.B).

A regular water quality-monitoring program should be implemented at Fluertys Creek to monitor all parameters, particularly faecal bacteria and turbidity levels.



**Table 4.23.B** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall |
|------------|-----------|--------------|-----------|-----------------|----------------|---------|
| Rating     | 3         | 2            | 4         | 4               | 2              | 15      |
| Assessment | Good      | Fair         | Excellent | Excellent       | Fair           | Good    |

## 4.14 Kermandie River

### Catchment description

Kermandie River (catchment area = 135 km<sup>2</sup>), in the western part of the catchment, drains off the hills surrounding Geeveston and travels over 12 km to flow into the Huon Estuary. The upper catchment is predominantly in State Forest, while the middle section of the catchment has been partially cleared for livestock grazing, orchards and rural residences.

Above Geeveston, Rileys Creek joins Kermandie River. Rileys Creek passes beside Geeveston Waste Disposal Area before entering a man-made reservoir, from which a controlled volume of water is allowed to continue down through grazing land to join the Kermandie River.

Within Geeveston, the river also collects water from two northern waterways, Scotts Rivulet and Crookes Rivulet. The catchments of both rivulets are largely forested, with lower sections cleared for orchards, livestock grazing and some rural residences. Below Geeveston, the Kermandie River receives effluent from the Geeveston STP, which services the immediate town area.

Between 1996 and 2000, extensive crack willow stands were removed from along Kermandie River, Rileys Creek, Scotts Rivulet and Crookes Rivulet by Kermandie Landcare. Significant replanting and fencing has only occurred in some areas.

### Monitoring program

Water temperature, turbidity, conductivity and pH were measured on 90 occasions at two sites in the upper catchment (Meyer and Lynch, 2000). Water temperature, turbidity, conductivity, pH and ortho-phosphate were monitored at eight sites during 1996-1997 (Table 4.14.A). Faecal bacteria levels were also measured on between two and four occasions at twelve sites along the catchment.

**Table 4.14.A** Kermandie River monitoring sites.

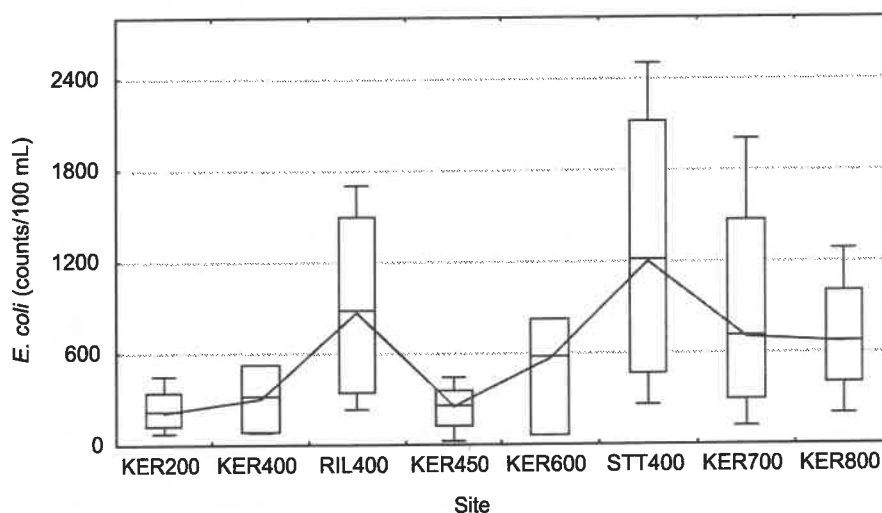
| Sites  | Site location                                   | Tested by                      | Times/year      | Start        | Finish          |
|--------|-------------------------------------------------|--------------------------------|-----------------|--------------|-----------------|
| KER001 | Bennetts Rd<br>(above Forestry Tasmania)        | Forestry Tasmania              | 26              | 1997         | Current         |
| KER002 | Kermandie River Rd<br>(below Forestry Tasmania) | Forestry Tasmania              | 26              | 1997         | Current         |
| KER200 | Kermandie River Rd at Golf Course               | Bobbi (1998)<br>Healthy Rivers | 2<br>Irregular  | 1997<br>1999 | 1997<br>Current |
| KER400 | Kermandie River Rd                              | Healthy Rivers                 | Irregular       | 1999         | Current         |
| RIL200 | Above dam                                       | Bobbi (1998)                   | 2               | 1997         | 1997            |
| RIL400 | Bottom of Rileys Creek Rd<br>(Rileys Creek)     | Bobbi (1998)<br>Healthy Rivers | 12<br>Irregular | 1996<br>1999 | 1997<br>Current |
| KER450 | RSL/Bowls Club                                  | Bobbi (1998)<br>Healthy Rivers | 2<br>Irregular  | 1997<br>1999 | 1997<br>Current |
| KER600 | Arve Rd                                         | Healthy Rivers                 | Irregular       | 1999         | Current         |
| STT100 | Donnellys Rd<br>(Scotts Rivulet)                | Bobbi (1998)                   | 2               | 1997         | 1997            |
| STT300 | Hawoods Rd<br>(Scotts Rivulet)                  | Healthy Rivers                 | Irregular       | 1999         | Current         |
| STT350 | Hank-Robertson Rd<br>(Crookes Rivulet)          | Healthy Rivers                 | Irregular       | 1999         | Current         |
| STT400 | Fourfoot Rd<br>(Scotts Rivulet)                 | Bobbi (1998)                   | 2               | 1997         | 1997            |
| KER700 | Spathfork Ranch                                 | Bobbi (1998)                   | 2               | 1997         | 1997            |
| KER800 | Below STP                                       | Bobbi (1998)                   | 12              | 1996         | 1997            |

### Results and Discussion

Meyer and Lynch (2000) results showed a flow of excellent quality water down through the catchment. Although the average turbidity increased from 0.88 NTU to 2.05 NTU between the upper and lower sites, the difference did not change the excellent water quality rating. While forestry activities do have the potential to effect water quality, the volume in Kermandie River may dilute any effects from increases in soil erosion, removal of vegetation and use of agricultural chemicals.

Below State Forest but above Rileys Creek, the water quality of Kermandie River in 1996/97 was good, with low conductivity, turbidity and ortho-phosphate levels, and slightly acidic pH levels. Conductivity, turbidity and ortho-phosphate levels were far higher in Scotts Rivulet and to a lesser degree in Rileys Creek (Bobbi, 1998).

Monitoring of *E. coli* during 1996-2000 indicated moderate levels in the Kermandie River from the monitoring site near the Golf Course right down to the bottom site, with Rileys Creek and Scotts Rivulet having very high levels (Fig. 4.14). Due in part to the bacterial rich water from Scotts Rivulet, the lower section Kermandie River also has high bacterial levels. However, faecal bacterial testing downstream of the Geeveston STP indicated that the bacterial loads from the STP might have reduced, with *E. coli* levels in 1999 significantly lower than results during 1997. This is likely to be due to improvements made to the aeration system of the STP in August 1998 (R. Bourke, personal communication).

**Fig. 4.14** *E. coli* levels along Kermandie River.

Water quality in Scotts Rivulet and Rileys Creek could be degraded due to a number of factors, including increased cattle access to the waterways following the removal of the crack willows and having some of the roadside drains along unsealed roads inputting directly into the adjacent waterways.

### Recommendations

Significantly more water quality data is required to form a complete picture about the health of the catchment. The limited water quality information suggests within the catchment there needs better management of domestic animals and septic systems near waterways, vegetation and other forms of soil disturbance and gravel road drainage (Table 4.14.B). Revegetation of riparian zones is required along much of the creek to improve shading.

Within the crack willow removal program, the current issues of shading, erosion and stock access in cleared areas need to be addressed before additional trees are removed.

A regular water quality-monitoring program should be implemented at Kermandie River, Rileys Creek and Scotts Rivulet to monitor all parameters.

**Table 4.14.B** Water quality assessment.

|                    | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall          |
|--------------------|-----------|--------------|-----------|-----------------|----------------|------------------|
| Kermandie R.       | 4         | 3            | 4         | Not monitored   | Not monitored  | <b>18</b>        |
| - within forest    | Excellent | Good         | Excellent |                 |                | <b>Excellent</b> |
| Rileys Creek       | 3         | 3            | 4         | 2               | 1              | <b>13</b>        |
|                    | Good      | Good         | Excellent | Fair            | Poor           | <b>Good</b>      |
| Scotts Rivulet     | 2         | 3            | 4         | 2               | 1              | <b>12</b>        |
|                    | Fair      | Good         | Excellent | Fair            | Poor           | <b>Good</b>      |
| Kermandie R.       | 3         | 3            | 4         | 2               | 1              | <b>13</b>        |
| - below Scotts Rvt | Good      | Good         | Excellent | Fair            | Poor           | <b>Good</b>      |

## 4.15 Lune River

### Catchment description

Lune River (catchment area = 142 km<sup>2</sup>) begins in the World Heritage Area southwest of Southport. Due to the numerous peaks and valleys in the catchment area, a variety of smaller creeks such as Moonlight Creek and Hot Springs Creek (Hastings Cave region) feed into Lune River. The catchment has formally and informally reserved land, timber harvesting and plantation areas. The small town of Lune River is found at the mouth of the river.

### Monitoring program

Water temperature, turbidity, conductivity and pH were tested on 90 occasions at two sites (Table 4.15.A).

**Table 4.15.A** Lune River monitoring sites.

| Sites  | Site location                                 | Tested by         | Times/year | Start | Finish  |
|--------|-----------------------------------------------|-------------------|------------|-------|---------|
| LUN001 | Hastings Cave Rd<br>(above Forestry Tasmania) | Forestry Tasmania | 26         | 1997  | Current |
| LUN002 | South Lune Rd<br>(below Forestry Tasmania)    | Forestry Tasmania | 26         | 1997  | Current |

### Results and Discussion

Meyer and Lynch (2000) results showed a flow of excellent quality water down through the catchment. While forestry activities do have the potential to effect water quality, the volume in Lune River may dilute any effects from increases in soil erosion, removal of vegetation and use of agricultural chemicals.

### Recommendations

The water quality information suggests that the catchment is not significantly effected by the present land uses and the current land and water management practices currently in place should be continued (Table 4.15.B). The current monitoring program should be continued. Agricultural chemical monitoring could be included if warranted.

**Table 4.15.B** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall   |
|------------|-----------|--------------|-----------|-----------------|----------------|-----------|
| Rating     | 4         | 3            | 4         | Not monitored   | Not monitored  | 18        |
| Assessment | Excellent | Good         | Excellent |                 |                | Excellent |

## 4.16 Prices Creek

### Catchment description

Prices Creek (catchment area = 7 km<sup>2</sup>), in the western catchment, runs off small foothills and travels only 4 km to the Huon Estuary. Much of the catchment is forested, with rural residences and a variety of agricultural activities in the middle and lower sections of the catchment. The riparian zone in the lower part of the catchment has an extensive stand of crack willow, which has blocked the flow and led to swampy conditions.

The township of Franklin, with 450 residents, is found near the mouth of the creek. In 1998, Franklin was connected to the Ranelagh-Huonville STP, significantly reducing the amount of residential effluent entering both Prices Creek and the Huon River.

### Monitoring program

Water temperature, turbidity, conductivity and pH were monitored on 20 occasions at three sites in the upper, middle and lower parts of the catchment (Table 4.16.A). Orthophosphate and *E. coli* levels were measured on two occasions at the bottom site.

**Table 4.16.A** Prices Creek monitoring sites.

| Sites  | Site location     | Tested by        | Times/year | Start | Finish  |
|--------|-------------------|------------------|------------|-------|---------|
| PCK010 | Rankins Rd        | Franklin Primary | 12         | 1998  | Current |
| PCK020 | Below Old Rd      | Franklin Primary | 12         | 1998  | Current |
| PCK030 | School footbridge | Bobbi (1998)     | 2          | 1997  | 1997    |
|        |                   | Franklin Primary | 12         | 1998  | Current |

### Results and Discussion

There was no difference in the median water temperature between the three sites, with all sites having a large seasonal range. The middle site, Site PCK020, had the lowest summer temperatures, which is probably due to an extensive and dense crack willow infestation above the site. (Fig. 4.16.A).

All three sites along Prices Creek had very low turbidity levels, with the median level at each site less than 7 NTU (Fig. 4.16.B). Elevated turbidity peaks (30 – 95 NTU) were recorded after heavy rain, with the maximum turbidity level recorded at each site decreasing down the catchment. This may be related to the higher levels of land and soil disturbance activities such as timber harvesting and Eucalyptus plantation establishment higher in the catchment. Other factors which may cause higher turbidity peaks in the upper and middle section of the catchment include poor gravel road drainage and the creek being forced to erode new paths due to the extensive willow stands.

Conductivity levels increased down the catchment, although the median conductivity at the bottom site of 218  $\mu\text{S}/\text{cm}$  still indicates fairly dilute water (Fig. 4.16.C). Median pH values are slightly acidic across the catchment, which is normal (Fig. 4.16.D).

Monitoring at the lower site in 1997 recorded elevated *E. coli* levels (above 100 counts/100 mL), with septic tanks being suggested as the key culprit (Bobbi, 1998). Although *E. coli* was not measured by Franklin Primary School's current monitoring program, levels may well be lower, due to the connection of Franklin to the Huonville-Ranelagh STP.

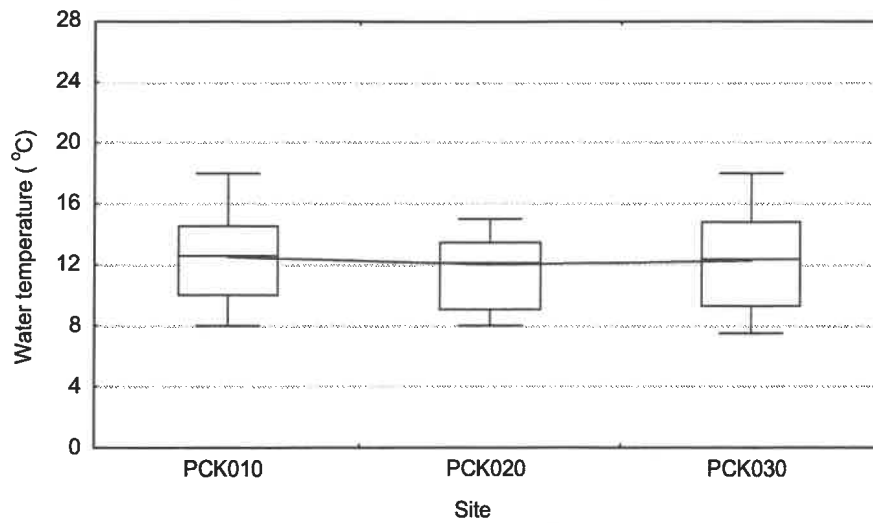
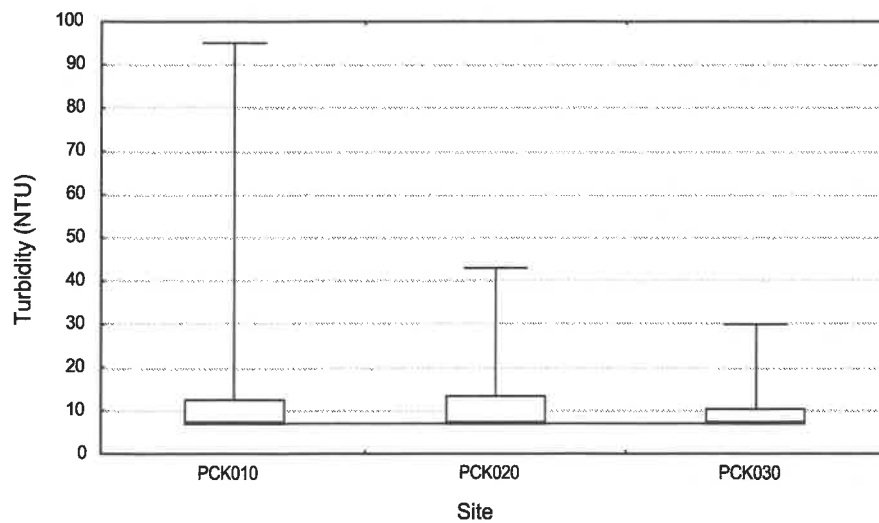
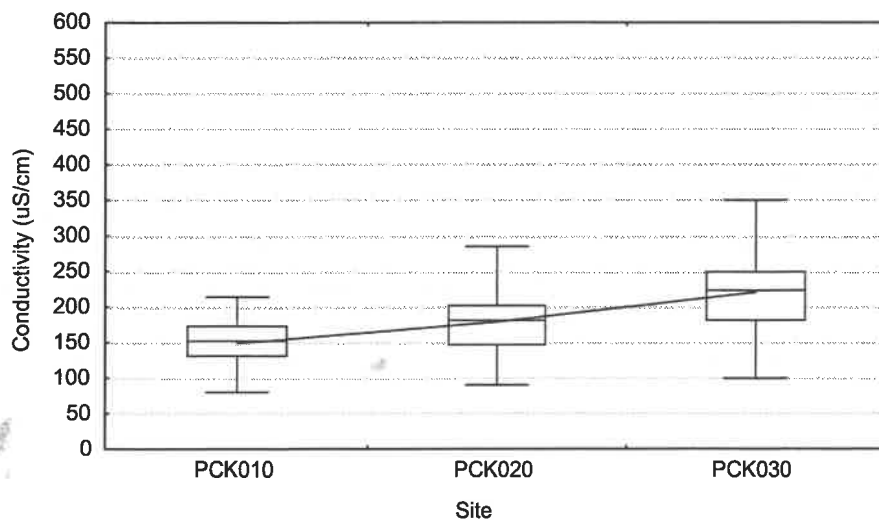
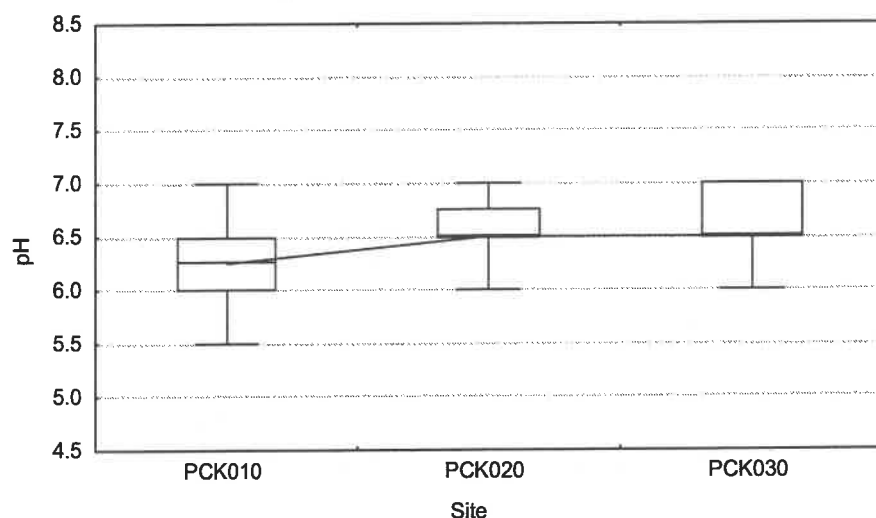
**Fig. 4.16.A Water temperature along Prices Creek.****Fig. 4.16.B Turbidity levels along Prices Creek.****Fig. 4.16.C Conductivity levels along Prices Creek.**

Fig. 4.16.D pH levels along Prices Creek



### Recommendations

The water quality information suggests within the catchment that there needs to be better management of septic systems and domestic animals near waterways, vegetation protection and minimising other forms of soil disturbance and gravel road drainage (Table 4.16.B). Rehabilitation of riparian zones is required along much of the creek to improve shading.

Crack willow trees should be removed where possible to reduce stream width and stream erosion during high flow. Before any willow removal, the issues of shading, erosion and stock access need to be addressed.

The monitoring program undertaken by Franklin Primary School should continue, with more regular *E. coli* and ortho-phosphate testing. The school should continue to receive the support of a Waterwatch network.

Table 4.16.B Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall |
|------------|-----------|--------------|-----------|-----------------|----------------|---------|
| Rating     | 4         | 3            | 4         | 3               | 2              | 16      |
| Assessment | Excellent | Good         | Excellent | Good            | Fair           | Good    |

## 4.17 Settlement Creek

### Catchment description

Settlement Creek is a small creek that winds its way down through five kilometres of forested and agricultural land to Southport Bay. Southport has a small permanent population and a summer time influx of visitors and shack owners. All residences have septic systems, with some older shacks close to the creek edge. In the middle of the catchment, there is a tip adjacent to the creek, although it has been a waste transfer station since 1994.

### Monitoring program

Two surveys in January and April 1999 measured water temperature, conductivity, pH and turbidity at five sites (Table 4.17.A).

**Table 4.17.A.** Settlement Creek monitoring sites.

| Sites  | Site location                      | Tested by            | Times/year | Start | Finish  |
|--------|------------------------------------|----------------------|------------|-------|---------|
| SOU010 | Huon Highway                       | Southport Waterwatch | Irregular  | 1999  | Current |
| SOU050 | Huon Hwy/Hastings Cave Rd junction | Southport Waterwatch | Irregular  | 1999  | Current |
| SOU100 | Above tip                          | Southport Waterwatch | Irregular  | 1999  | Current |
| SOU150 | Above pub                          | Southport Waterwatch | Irregular  | 1999  | Current |
| SOU200 | Opposite 'Jetty House'             | Southport Waterwatch | Irregular  | 1999  | Current |

### Results and Discussion

**Table 4.17.B** Water quality results along Settlement Creek.

| Date     | Site   | Water temperature (°C) | Turbidity (NTU) | Conductivity (µS/cm) | pH  |
|----------|--------|------------------------|-----------------|----------------------|-----|
| 28/01/99 | SOU100 | 16                     | 10              | 740                  | 7   |
|          | SOU150 | 19                     | 20              | 530                  | 7   |
|          | SOU200 | 23                     | <7              | 1000                 | 8.5 |
| 1/04/99  | SOU010 | 10                     | 16              | 210                  | 7   |
|          | SOU050 | 10                     | 20              | 260                  | 6   |
|          | SOU100 | 12                     | 14              | 340                  | 7   |
|          | SOU150 | 12                     | 12              | 340                  | 6.5 |
|          | SOU200 | 14                     | 15              | 12800                | 7.5 |

The conductivity results indicated that the lowest site was within the tidal limit of Southport Bay (Table 4.17.B). Conductivity levels in the creek were also moderately high and increased down the catchment. The slightly acidic pH level in the creek and slightly alkaline pH level in the estuary were within normal ranges (Table 4.17.B). Warm water temperatures were recorded in both January and April, with the range between 10 and 23 °C (Table 4.17.B). These are somewhat elevated results given the creek has reasonably vegetated riparian zones. Turbidity levels were between less than 7 and 20 NTU, which are above the normal levels for undisturbed waterways (Table 4.17.B).

### Recommendations

Significantly more water quality data is required to form a complete picture about the health of the catchment. The limited information suggests within the catchment there needs to be better management of domestic animals near waterways and vegetation management and other forms of soil disturbance (Table 4.17.C).

A regular water quality monitoring program should be implemented at Settlement Creek to monitor all parameters, including faecal bacteria.

**Table 4.17.C.** Water quality assessment.

| Parameter  | Turbidity | Conductivity | pH        | Ortho-phosphate | <i>E. coli</i> | Overall     |
|------------|-----------|--------------|-----------|-----------------|----------------|-------------|
| Rating     | 3         | 2            | 4         | 2               | Not monitored  | <b>13</b>   |
| Assessment | Good      | Fair         | Excellent | Fair            |                | <b>Fair</b> |



## 4.18 Agnes Rivulet

### Catchment description

Agnes Rivulet (catchment area = 43 km<sup>2</sup>) travels 8 km from its source on Grey Mountain to Port Cygnet Bay. Two significant side creeks drain into Agnes Rivulet: Supplices Creek, which drains into Agnes Rivulet above Cygnet and 'Tip' Creek, which drains in 300 m upstream of the mouth.

While the upper catchment of Agnes Rivulet is predominantly privately owned Eucalyptus forest, the mid-catchment has been cleared and has many agricultural activities including livestock grazing, orchards and rural residences. The extensive streamside crack willow stands were removed by Port Cygnet Landcare in 1998 and the riparian zone was largely replanted with native trees, but not fenced.

Supplices Creek enters Agnes Rivulet at the junction of Channel Highway and Slab Road (Fig. 4.18). It is a small creek that drains from the steep hills north west of Cygnet. Much of the catchment of Supplices Creek has been cleared for orchards, livestock grazing, a dairy and rural residences. For irrigation purposes, there are a number of in-stream dams and other water off-take sites. The creek was heavily infested with crack willows, which were blocking and spreading the flow of the creek. These willows were removed in 1999 and along most of the creek the riparian zone was fenced and replanted with native species.

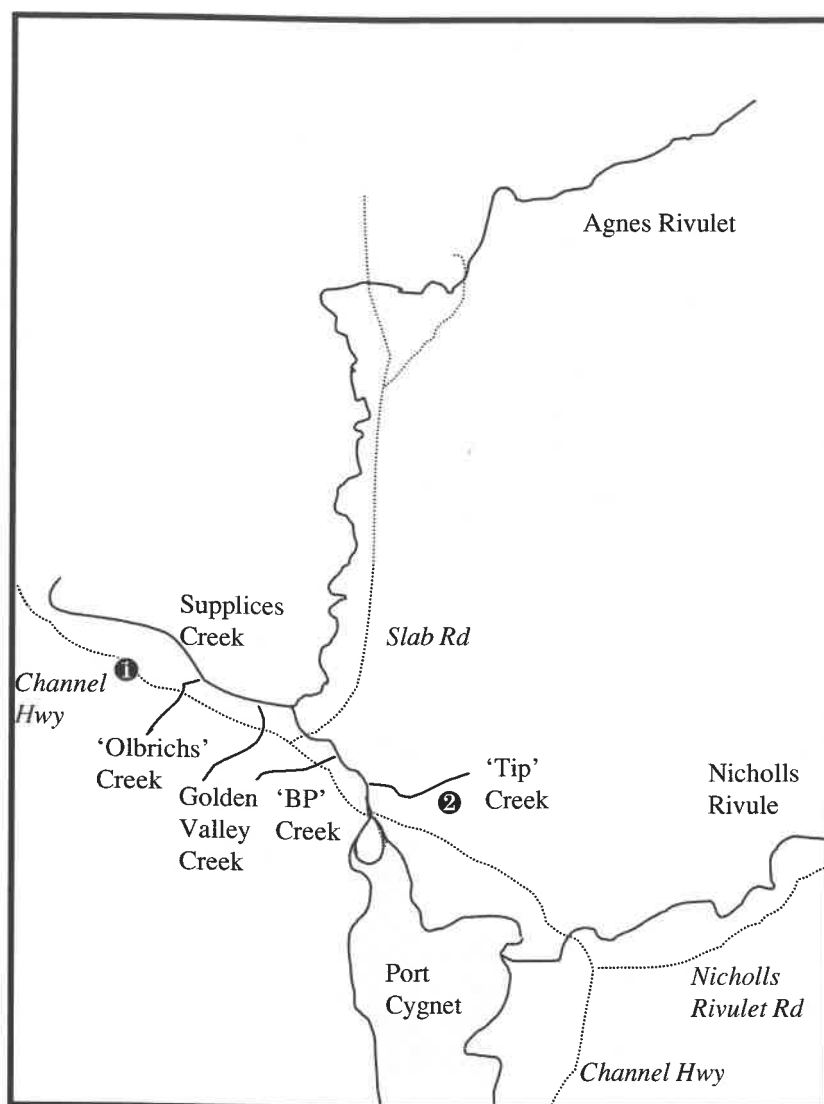
'Olbrichs' Creek is a small drainage creek, which enters 200 m downstream of Stevens Road. Supplices Creek is also joined near its mouth by Golden Valley Creek (Fig. 4.18). The catchment of Golden Valley Creek is a mix of forest and cleared land for rural residences, orchards, and livestock grazing. The catchment area is quite small and consequently, the creek is fed predominantly by groundwater.

The township of Cygnet, with its urban dwellings, roads and other infrastructure for 850 residents dominates the lower catchment area of Agnes Rivulet. Cygnet has a STP, although the connections are limited to the central urban area only. The STP effluent is piped into Port Cygnet.

In the western part of the town, 'BP' Creek and stormwater are piped together underground to avoid homes and the creek emerges near the BP Service Station on Channel Highway (Fig. 4.18). The small volume in 'BP' Creek flows a further 250 m to enter Agnes Rivulet.

Just east of the Cygnet township is another significant side creek, 'Tip' Creek. Although small, 'Tip' Creek is monitored because it flows past the Cygnet Waste Disposal Area and the Cygnet Golf Course, before entering Agnes Rivulet near the Cygnet Town Hall (Fig. 4.18).

The lower section of Agnes Rivulet drains through very dense streamside crack willows and the rivulet splits into two as it separates around the Burtons Reserve wetlands before entering Port Cygnet.



**Fig. 4.18** Agnes Rivulet and location of subcatchment waterways.  
Scale = 1: 80 000

|             |                                            |
|-------------|--------------------------------------------|
| <b>Key:</b> |                                            |
|             | Waterway                                   |
|             | Road                                       |
| ①           | Dairy along Supplies Creek                 |
| ②           | Cygnnet Waste Disposal Area al 'Tip' Creek |

### Monitoring program

Water temperature, turbidity, conductivity and pH were measured on between ten and twenty-seven occasions at seven sites along Agnes Rivulet. *E. coli* levels were measured on two occasions at an upper site (AGN016) and a lower site (AGN130) in 1997 (Table 4.18.A).

In Supplices Creek, water temperature, turbidity, conductivity, pH, faecal bacteria, dissolved oxygen and ortho-phosphate were measured on between nine and twelve occasions at seven sites (Table 4.18.A). The bottom site at Golden Valley Creek was monitored for water temperature, turbidity, conductivity and pH on ten occasions. *E. coli* levels were measured on two occasions at the bottom sites in both creeks in 1997.

Water temperature, turbidity, conductivity, pH and *E. coli* were measured in the canal near the BP Service Station on between five and eleven occasions. 'Tip' Creek had four monitoring sites and water temperature, turbidity, conductivity and pH were measured on between six and ten occasions (Table 4.18.A).

**Table 4.18.A** Agnes Rivulet monitoring sites.

| Sites  | Site location                       | Tested by               | Times/year | Start | Finish  |
|--------|-------------------------------------|-------------------------|------------|-------|---------|
| AGN010 | Above town water dam                | Bobbi (1998)            | 2          | 1997  | 1997    |
|        |                                     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN016 | Woodcock Rd                         | Bobbi (1998)            | 2          | 1997  | 1997    |
|        |                                     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN120 | Football Oval                       | Cygnnet Junior Landcare | 4          | 1998  | Current |
| SUP010 | Supplices Rd - Supplices Ck         | Port Cygnnet Watercare  | 12         | 1999  | Current |
| SUP020 | Below underpass - Supplices Ck      | Port Cygnnet Watercare  | 12         | 1999  | Current |
| SUP050 | Pop's fruit shop - Supplices Ck     | Cygnnet Junior Landcare | 4          | 1998  | 1999    |
|        |                                     | Port Cygnnet Watercare  | 12         | 1999  | Current |
| SUP100 | Stevens Rd - Supplices Ck           | Port Cygnnet Watercare  | 12         | 1999  | Current |
| SUP200 | Below pottery - 'Olbrichs' Ck       | Cygnnet Junior Landcare | 4          | 1998  | 1997    |
|        |                                     | Port Cygnnet Watercare  | 12         | 1999  | Current |
| SUP300 | U/S Golden Valley Ck - Supplices Ck | Bobbi (1998)            | 2          | 1997  | 1997    |
|        |                                     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| GVC115 | Dances Rd - Golden Valley Ck        | Bobbi (1998)            | 2          | 1997  | 1997    |
|        |                                     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| SUP350 | D/S Golden Valley Ck - Supplices Ck | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN130 | Scout Hall                          | Bobbi (1998)            | 12         | 1996  | 1997    |
|        |                                     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN200 | Service station canal - 'BP' Ck     | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN245 | Dam above tip - 'Tip' Ck            | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN249 | Tip settlement pond                 | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN250 | Below tip - 'Tip' Ck                | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN260 | Golf Course - 'Tip' Ck              | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN300 | Town Hall                           | Cygnnet Junior Landcare | 4          | 1998  | Current |
| AGN400 | Channel Hwy                         | Cygnnet Junior Landcare | 4          | 1998  | Current |

## Results and Discussion

### 4.18.1 Supplices Creek

The median water temperature at the eight sites was around 12 °C. Very warm summer temperatures were experienced, with the maximum temperatures at all sites above 22 °C (Fig. 4.18.1.A). The warm water temperatures resulted in low levels of dissolved oxygen

around 8 mg/L and the percent saturation levels at seven of the sites were only just above 80 % (Fig. 4.18.B and C). The site below the dairy had a median dissolved oxygen percent saturation of 73 %, indicating very low levels would occur during the night.

Turbidity levels in the catchment were very elevated with six sites having median values around 30 NTU and the site below the dairy having a median level of 85 NTU and 'Olbrichs' Creek recording a median level of 65 NTU (Fig. 4.18.1.D). In an unaltered state, Supplices Creek would normally have had turbidity levels around 5 to 10 NTU.

The high level of suspended sediment has also resulted in high dissolved solutes, with median conductivity levels close to 600  $\mu\text{S}/\text{cm}$  (Fig. 4.18.1.E). 'Olbrichs' Creek in contrast, had a lower median conductivity level at 325  $\mu\text{S}/\text{cm}$ . Golden Valley Creek experienced very high conductivity levels during low flows, due to high dominance of groundwater compared to surface and sub-surface flows.

Median pH levels at the eight sites were slightly acidic at between 6.0 and 6.5, which is normal (Fig. 4.18.1.F).

Median ortho-phosphate levels at the seven sites in Supplices Creek were between 0.045 and 0.28 mg/L (Fig. 4.18.1.G), which is very elevated compared to normal natural background levels less than 0.008 mg/L. The site below the dairy appeared to be a significant source of ortho-phosphate, this needs to be confirmed with a monitoring site above the farm operations. Golden Valley Creek in comparison had a median level of 0.011 mg/L.

*E. coli* monitoring in 1997 recorded levels above 400 counts/100 mL in the bottom sites of both Supplices Creek (SUP300) and Golden Valley Creek (GVC115) (Bobbi, 1998).

Fig. 4.18.1.A Water temperature along Supplices Creek.

